Planning and action selection

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Planning and action selection before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your ex**plan**ation.

State machines for bounded agents

Without opening the old lesson, explain one key idea from State machines for bounded agents. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Failure budgets and service objectives

Without opening the old lesson, explain one key idea from Failure budgets and service objectives. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Testing model-dependent behaviour

Without opening the old lesson, explain one key idea from Testing model-dependent behaviour. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Lesson reference for workbook practice

Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

Planning proposes steps; execution must still validate permissions and **prerequisite**s for each step. Re**plan**ning cannot invent authority.

For read-only reporting, allowed **plan** is validate -> retrieve -> verify -> answer. If retrieval lacks **evidence**, route to review. Reject write or send steps before they run.

Practise with fading help

> Complete the supported Planning and action selection trace and record the first step you can explain without help.

Use W2: Trace the worked case for **Plan**ning and **action selection**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

Individual reasoning check

> Check the proposed Planning and action selection explanation against the supplied facts before discussing it.

Use the worked case for **Plan**ning and **action selection**. Proposed ex**plan**ation: “The worked result establishes every future case”. Decide whether this ex**plan**ation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 15

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply a new case independently

Plan validates **input** then sends a message before checking recipient approval. The task authorises drafting only. Locate violation.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Explain the mechanism

> Explain the mechanism. Apply the supplied Planning and action selection facts and keep your reasoning.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

W2 Trace the worked case

> Trace the worked case. Apply the supplied Planning and action selection facts and keep your reasoning.

Generated **plan** retrieves a room, books it and then answers. It calls booking necessary for helpfulness. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

W3 Audit the limit

> Audit the limit. Apply the supplied Planning and action selection facts and keep your reasoning.

A report omits this qualification: A plausible **plan** is not an approval or proof that its **prerequisite**s hold. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

> **My reasoning and evidence**

W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Planning and action selection facts and keep your reasoning.

Plan validates **input** then sends a message before checking recipient approval. The task authorises drafting only. Locate violation.

> **My reasoning and evidence**

WORKBOOK · 9 OF 15

W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied Planning and action selection facts and keep your reasoning.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Planning and action selection to Grounded Learning Service, identify a limit and choose the next useful test.

Add a labelled **plan**ning and **action selection evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of **plan**ning and **action selection**.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 12 OF 15

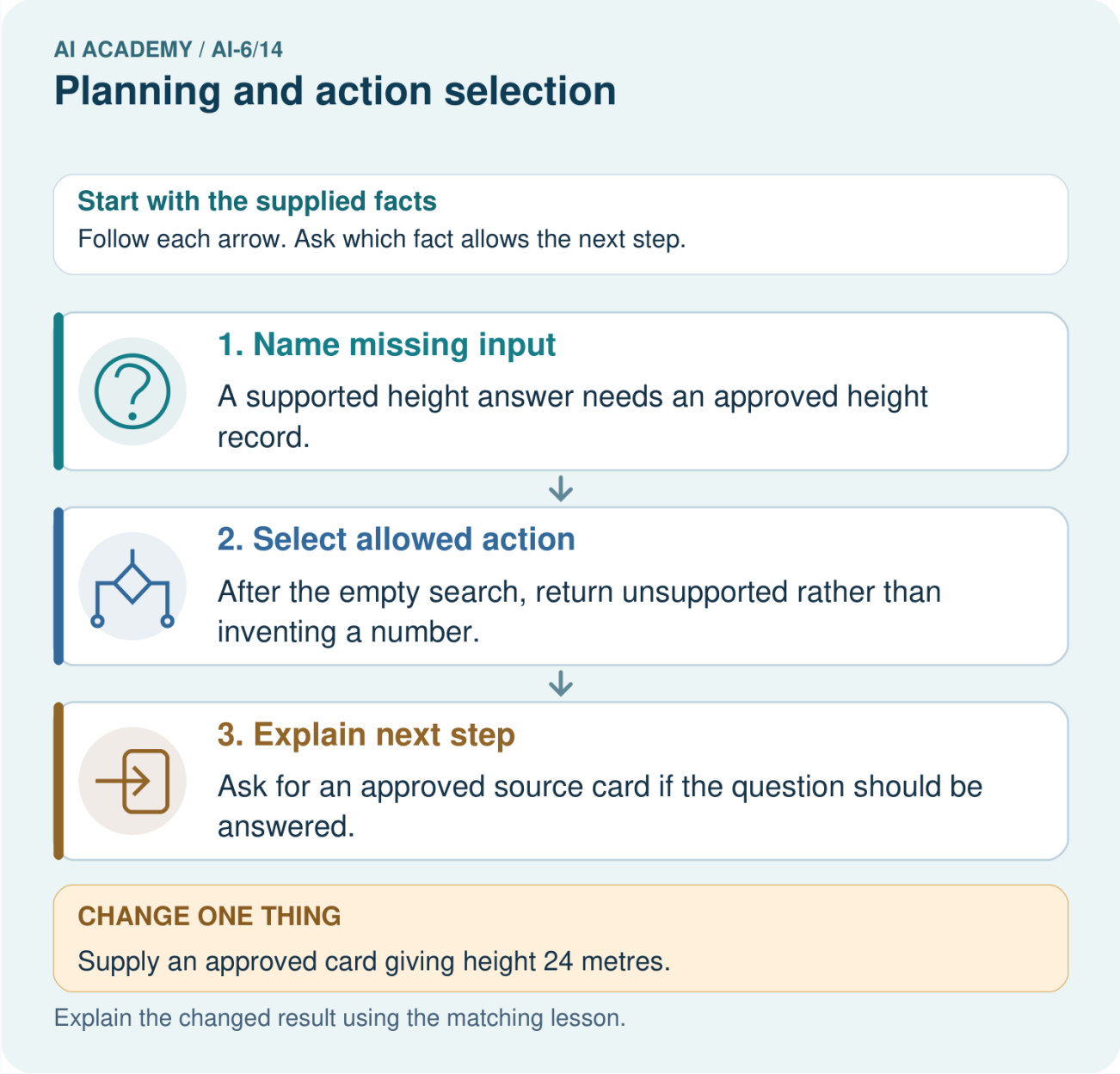
Pictorial case practice: Planning and action selection

> Plan around missing evidence. A bot must answer a tower-height question using approved

> cards only. Explain the labelled step before checking the changed condition.

Plan around missing **evidence**. A bot must answer a tower-height question using approved cards only.

Search finds no height card. Allowed actions: search, clarify, return unsupported. Unapproved web access is excluded.



Follow the labelled picture with the matching explanation. Open the vector diagram

Choose a permitted **plan** when no approved tower-height card exists. What new answer becomes supported after a card records 24 metres, and why could the bot not invent it earlier?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

WORKBOOK · 13 OF 15

Project portfolio milestone

Keep this week's input, deciding step, result and limit together in the Grounded Learning Service record.

Project: Grounded Learning Service

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Plan around missing **evidence**. A bot must answer a tower-height question using approved cards only.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Ask for an approved source card if the question should be answered.

1. Select allowed action

2. Explain next step

3. Name missing **input**

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported ex**plan**ation: Explain next step Ask for an approved source card if the question should be answered. **Plan**ning selects useful actions within available **evidence** and authority.

More steps cannot create permission to access an excluded source.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Supply an approved card giving height 24 metres. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. The bot can retrieve it and answer 24 metres with attribution.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported ex**plan**ation: The bot can retrieve it and answer 24 metres with attribution. The missing factual **input** is now available within the allowed collection.

Repair a missing building block

Review State machines for bounded agents (ai-6/13). Goal: Explain the **mechanism** and terminology of state machines for bounded agents. Short practice: Trace A and B through the state diagram.

Name B's forbidden transition and explain whether adding checking→querying permits that shortcut.. Return to this week and explain what you changed.

Review Failure budgets and service objectives (ai-6/11). Goal: Explain the **mechanism** and terminology of failure budgets and service objectives. Short practice: Calculate the maximum allowed failures among 100 requests.

Does 97 successes and 3 failures meet the target, and what changes after one failure becomes a success?. Return to this week and explain what you changed.

Review Testing model-dependent behaviour (ai-6/8). Goal: Explain the **mechanism** and terminology of testing model-dependent behaviour.

Short practice: Score "Friday in room B" and "Friday in room D" against both mandatory criteria. Would a correct 25-word response pass, and why?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

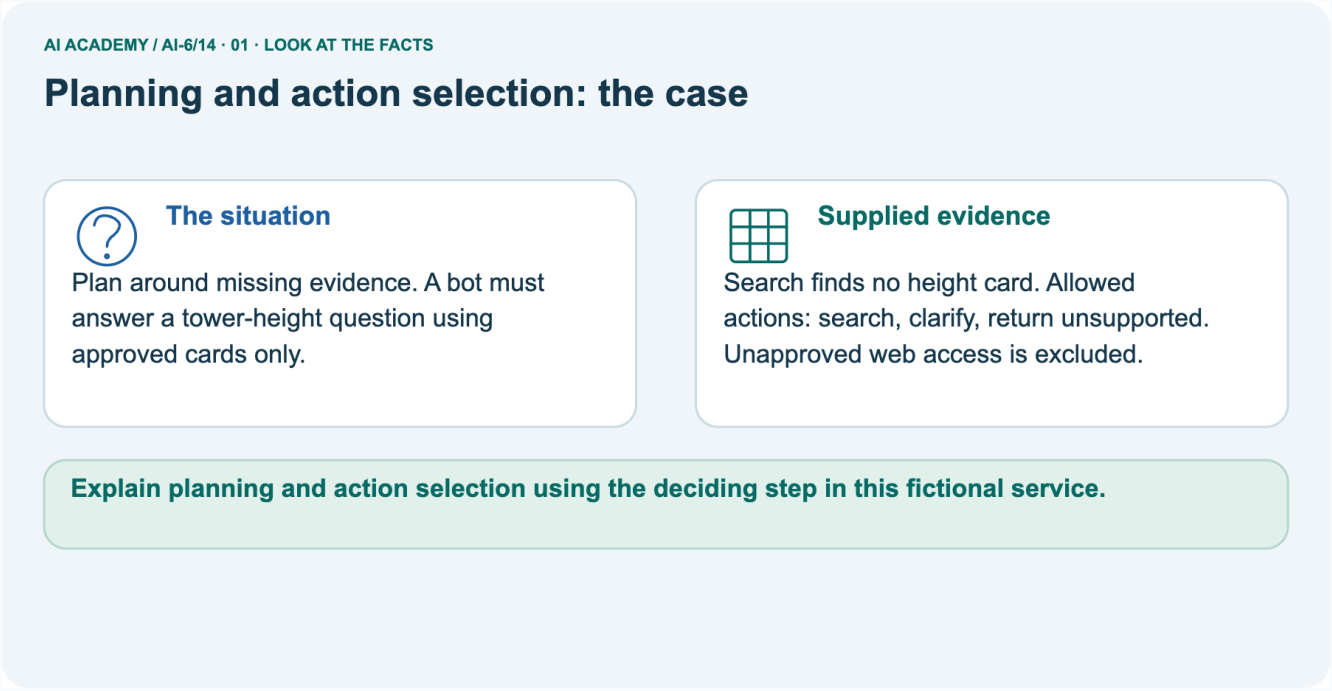
What I repaired, the evidence I kept and the next test

WORKBOOK · 15 OF 15

Visual concept journey

> Explain each Planning and action selection picture from its labels, then record a relevant limit.

01. Planning and action selection: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

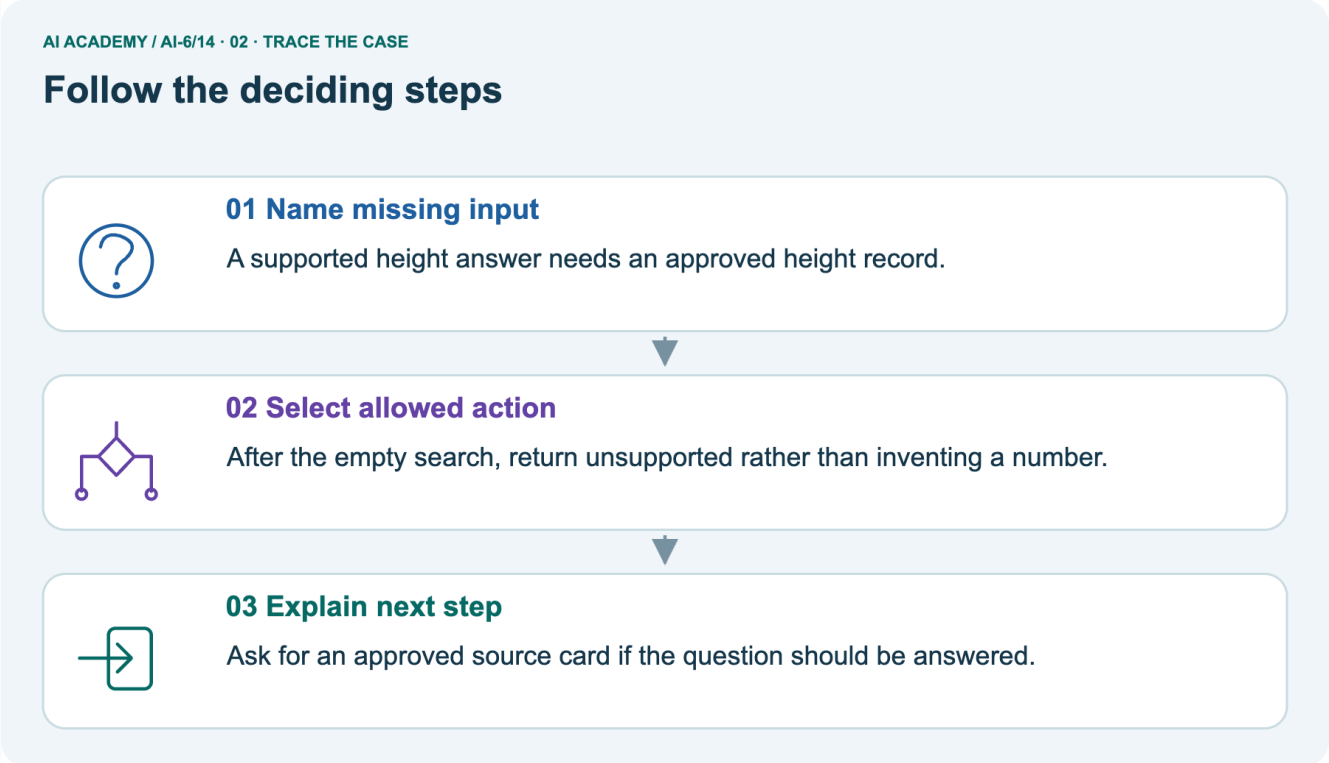
Plan around missing **evidence**. A bot must answer a tower-height question using approved cards only. Supplied facts: Search finds no height card. Allowed actions: search, clarify, return unsupported.

Unapproved web access is excluded.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



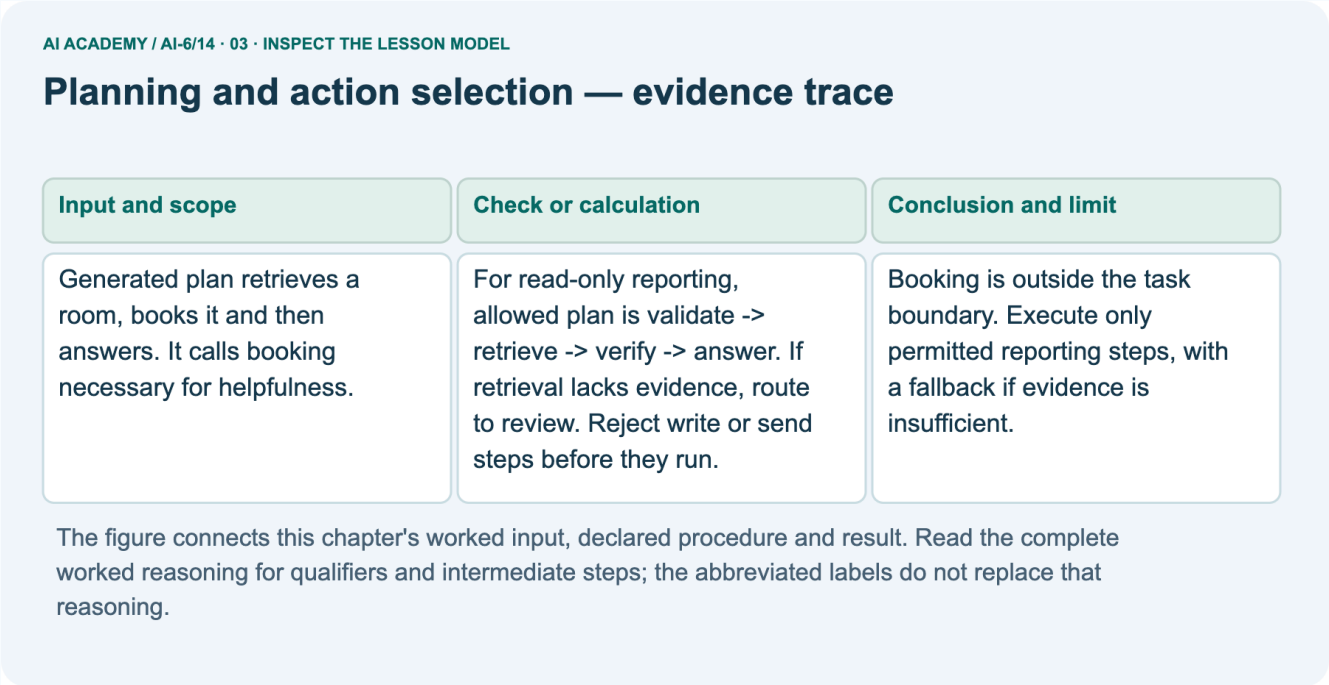
Follow the labelled picture with the matching explanation. Open the vector diagram

Name missing **input**: A supported height answer needs an approved height record. → Select allowed action: After the empty search, return unsupported rather than inventing a number. → Explain next step: Ask for an approved source card if the question should be answered.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Planning and action selection — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Planning and **action selection** — **evidence** trace. Generated **plan** retrieves a room, books it and then answers.

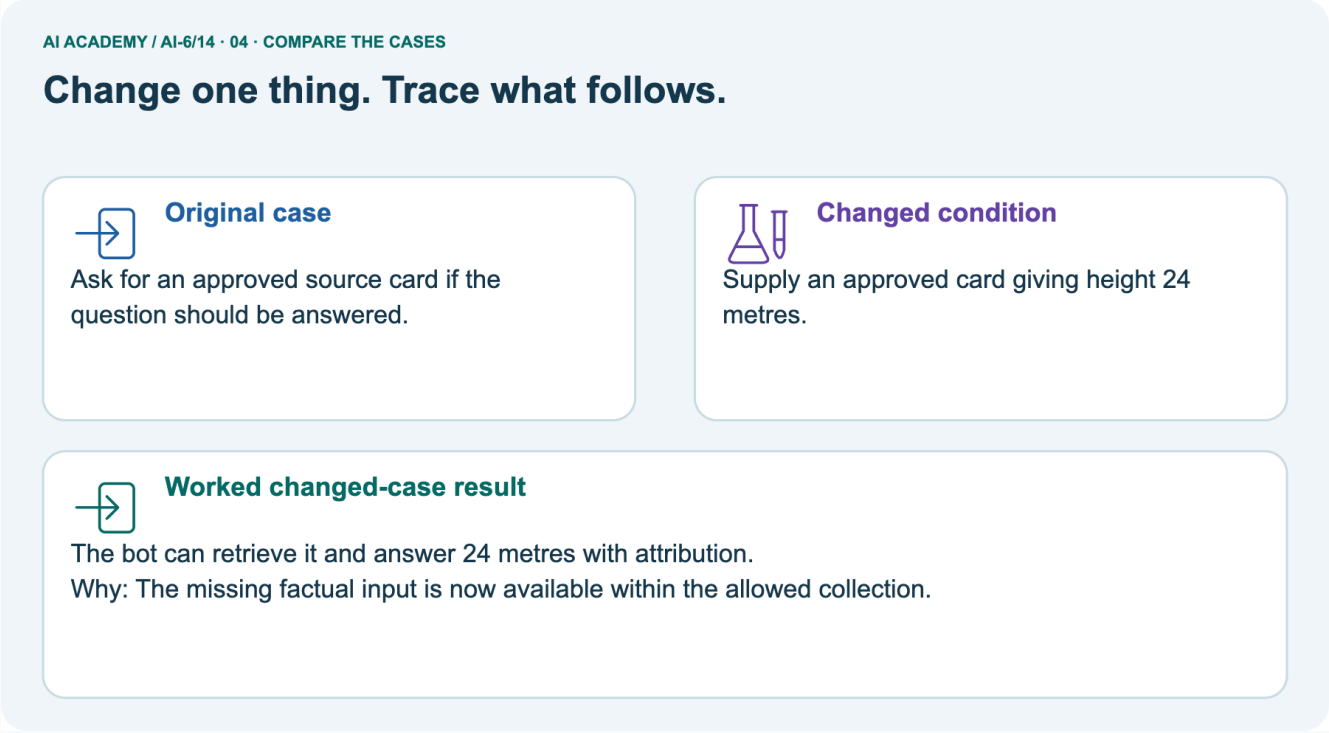
It calls booking necessary for helpfulness. | For read-only reporting, allowed **plan** is validate -> retrieve -> verify -> answer. If retrieval lacks **evidence**, route to review.

Reject write or send steps before they run. | Booking is outside the task boundary. Execute only permitted reporting steps, with a fallback if **evidence** is insufficient.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

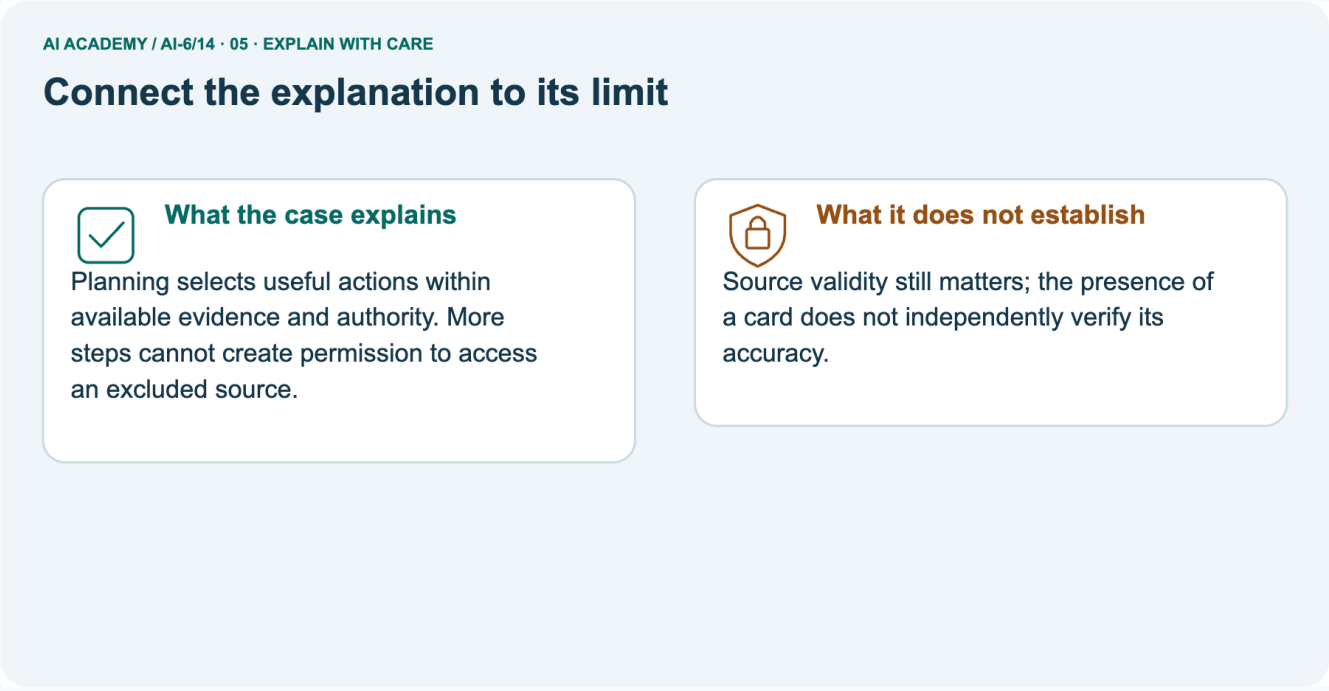
Original case: Ask for an approved source card if the question should be answered. Changed condition: Supply an approved card giving height 24 metres.

Expected result: The bot can retrieve it and answer 24 metres with attribution. Reason: The missing factual **input** is now available within the allowed collection.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

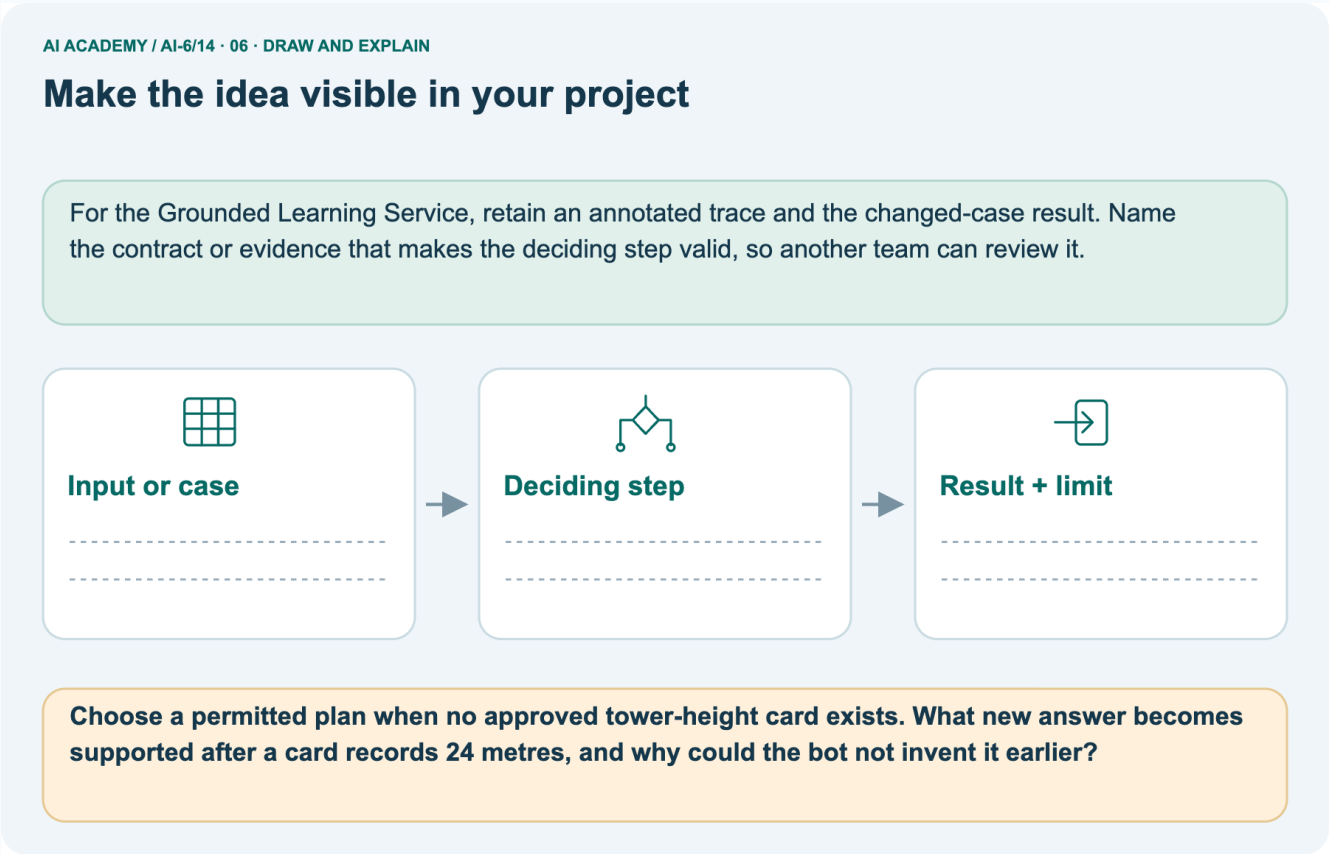
Planning selects useful actions within available **evidence** and authority. More steps cannot create permission to access an excluded source.

Limit: Source validity still matters; the presence of a card does not independently verify its accuracy.

> Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Choose a permitted **plan** when no approved tower-height card exists.

What new answer becomes supported after a card records 24 metres, and why could the bot not invent it earlier?

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.